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Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Hannah Kettle MATHS TUTORIALS

Half Past Paper – Week 19

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
1a	Using a written method to multiply decimals	3	U293
1b	Using a written method to divide with decimals	3	U868
3	Adding and subtracting mixed numbers	3	U793
5	Sharing amounts in a given ratio	3	U577
7	Finding the surface area of spheres, Finding the surface area of cubes and cuboids	4	U893, U929
9a	Calculating with roots and powers	1	U851
9b	Index rules with negative indices	1	U694
9c	Index rules with negative indices	1	U694
9d	Indices of the form 1/a	1	U985
11	Problem solving: Area and perimeter, Changing the subjects of formulae	4	U934, U556
13	Converting recurring decimals to fractions	1	U689
15	Adding and subtracting algebraic fractions	3	U685
17	Drawing graphs of inequalities	3	U747
19	Rationalising denominators containing two terms	4	U281
21a	Substituting into composite functions	2	U895
21b	Finding composite functions, Finding inverse functions	4	U448, U996
	Total	41	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 (*a*) Work out 3.67×4.2

.....

(3)

(*b*) Work out $59.84 \div 1.6$

.....

(3)

(Total for Question 1 is 6 marks)

3 Work out $4\frac{1}{5} - 2\frac{2}{3}$

Give your answer as a mixed number.

.....

(Total for Question 3 is 3 marks)

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5 Rosie, Matilda and Ibrahim collect stickers.

number of stickers Rosie has : number of stickers Matilda has : number of stickers Ibrahim has = 4 : 7 : 15

Ibrahim has 24 more stickers than Matilda.

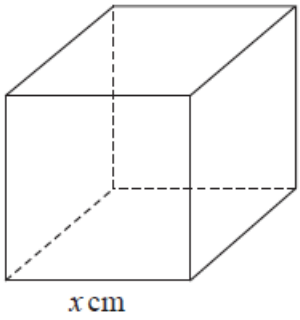
Ibrahim has more stickers than Rosie.
How many more?

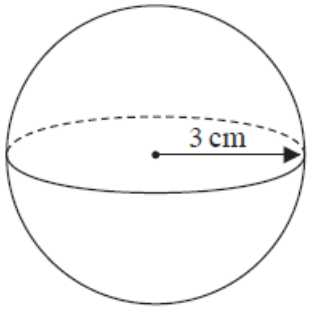
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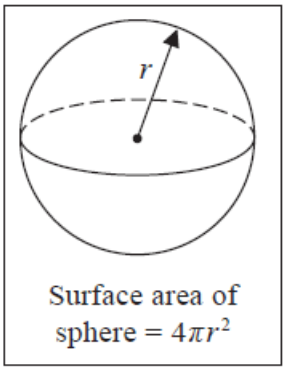
(Total for Question 5 is 3 marks)

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7 The diagram shows a cube with edges of length x cm and a sphere of radius 3 cm.







The surface area of the cube is equal to the surface area of the sphere.

Show that $x = \sqrt{k\pi}$ where k is an integer.

(Total for Question 7 is 4 marks)

9 (*a*) Write down the value of 7^0

.....

(1)

(*b*) Find the value of $3 \times 3^6 \times 3^{-6}$

.....

(1)

(*c*) Find the value of 2^{-4}

.....

(1)

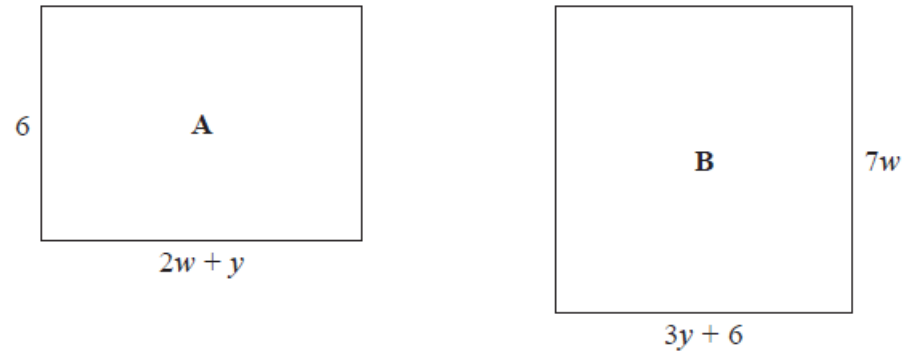
(*d*) Find the value of $27^{\frac{1}{3}}$

.....

(1)

(Total for Question 9 is 4 marks)

11 The diagram shows two rectangles, **A** and **B**.



All measurements are in centimetres.

The area of rectangle **A** is equal to the area of rectangle **B**.

Find an expression for y in terms of w.

.....

(Total for Question 11 is 4 marks)

13 Ted is trying to change $0.\dot{4}\dot{3}$ to a fraction.

Here is the start of his method.

$$x = 0.\dot{4}\dot{3}$$
$$10x = 4.\dot{3}\dot{4}$$
$$10x - x = 4.\dot{3}\dot{4} - 0.\dot{4}\dot{3}$$

Evaluate Ted's method so far.

.....

.....

.....

(Total for Question 13 is 1 mark)

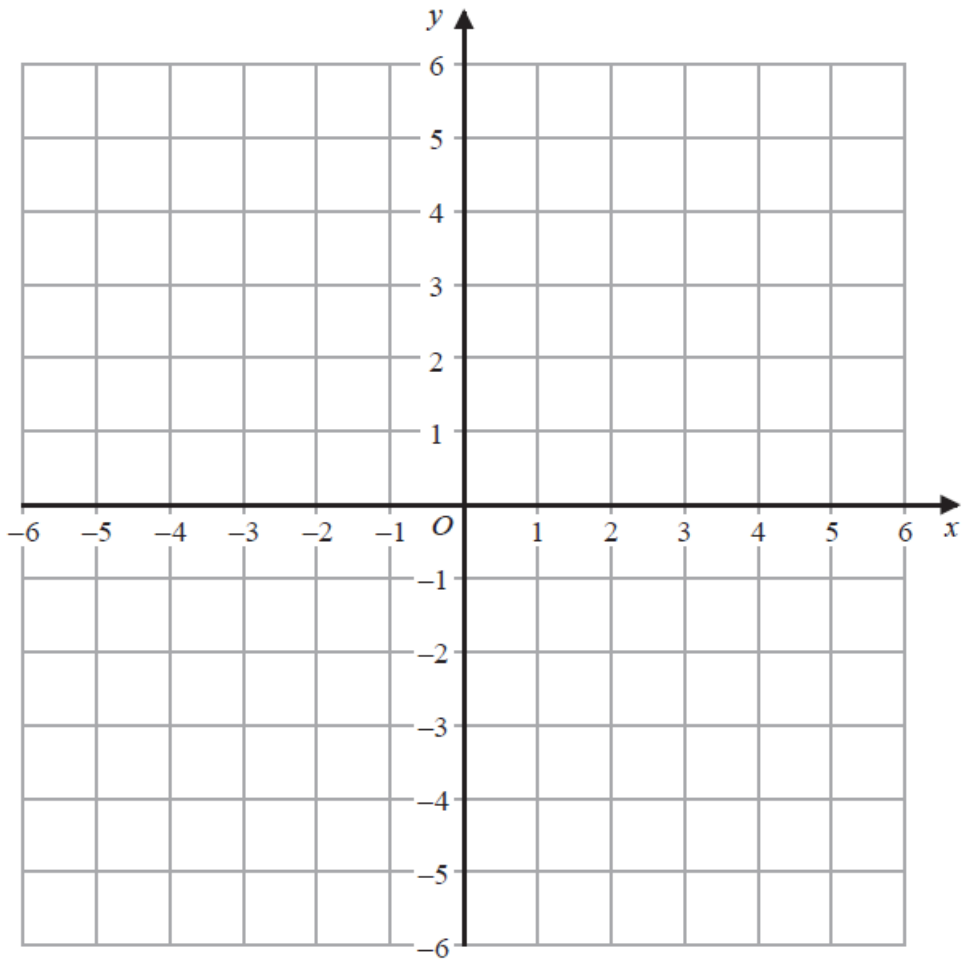
15 Show that $\frac{4x+3}{2x}+\frac{3}{5}$ can be written in the form $\frac{ax+b}{cx}$ where a, b and c are integers.

(Total for Question 15 is 3 marks)

17 On the grid show, by shading, the region that satisfies all of these inequalities.

$$2y + 4 < x \qquad x < 3 \qquad y < 6 - 3x$$

Label the region **R**.



(Total for Question 17 is 3 marks)

19 Show that $\frac{8+\sqrt{12}}{5+\sqrt{3}}$ can be written in the form $\frac{a+\sqrt{3}}{b}$, where a and b are integers.

(Total for Question 19 is 4 marks)

21 The functions f and g are such that

$$f(x) = 3x^2 + 1 \text{ for } x > 0 \qquad \text{and} \qquad g(x) = \frac{4}{x^2} \text{ for } x > 0$$

(*a*) Work out gf(1)

.....

(2)

The function h is such that $h = (fg)^{-1}$

(*b*) Find h(*x*)

.....

(4)

(Total for Question 21 is 6 marks)

TOTAL FOR PAPER IS 41 MARKS